



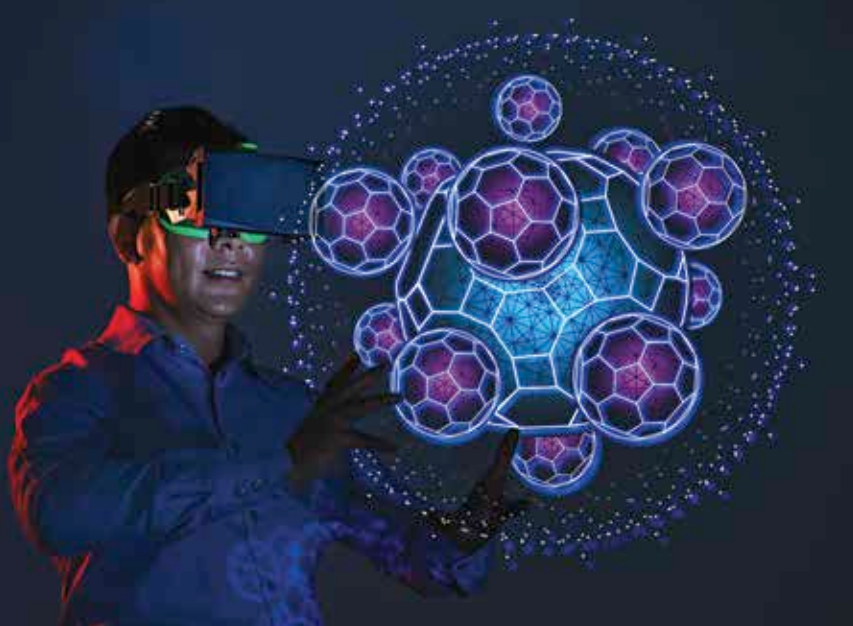
Quasar Tsunamis

The massive black holes at the centre of some galaxies are tearing them up from the inside. <https://digit.in/apr20-09>



SpaceX is hiring

Amidst all the panic surrounding COVID-19, SpaceX still wants to send Starship to Mars. <https://digit.in/apr20-10>



REACHING NEW HEIGHTS WITH

XR

Scientists are harnessing the power of XR for new developments and innovations

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he ideas of VR (Virtual Reality), MR (Mixed Reality) and AR (Augmented Reality) are ones that are famil-

iar to most people by now. However, it is not uncommon for people to perceive these technologies are mostly useful for entertainment. In reality, these technologies, collectively termed as XR (Extended Reality), hold massive potential in various fields of science, be it innovation, development, or education. The ideas have already begun to be brought to

fruition by scientists all over with interesting applications such as VR being used as a training ground for medical students and surgeons, AR being used in microscopes (developed by Google) which are able to detect cancerous cells in real-time, and other such novel applications. In 2018, a 'Hackathon' was conducted by Springer Nature, where participants used AR/VR headsets and accessed millions of data points through multiple APIs. The teams then attempted to solve real world problems by visualising data in AR/VR across scientific fields. Programs such as these, and other innovations/developments aided by XR, show that the technology can help scientists progress in their fields by providing enhanced immersiveness and context.

IN THE LABORATORY

The realm of mixed reality is promising to bring a new age of innovation in scientific laboratories across the world. For the uninitiated, mixed reality (MR)

is a technology that attempts to remove the boundaries between real and virtual interactions. This is done via occlusion where computer-generated objects and real physical objects co-exist and interact in real-time. It is a combination of VR and AR essentially. By using Mixed Reality, users can control the physical objects by using hand gestures, voice commands, and even eye movements. Therefore, scientists in laboratories can manipulate their lab benches without so much as touching it.

Holo4Labs attempts to revolutionise the laboratory by integrating MR into this space. Developed by Solution4Labs, Holo4Labs is the mixed reality variant of Laboratory Information Manage-

ment Software (LIMS) made specifically for Microsoft HoloLens 2. Scientists are able to merge data processing, information management software with the day-to-day ongoing of a lab. The goal of Holo4Labs is to simplify data management in labs and streamline operations. Think of the technology as an extra set of hands for scientists, however, since it is computer-based, the room for human error is reduced. One of the main goals of Holo4Labs is to free up scientists from having to do mundane tasks in the lab and be more focused on innovation.

Holo4Labs runs Thermo Fisher's Scientific SampleManager LIMS, due to which the mixed reality system can tap into laboratory-specific applications. This is especially true in industries such as fuel, pharmaceuticals, food, biobanking and metallurgy since they are already well-integrated with the software. As compared to traditional AR, mixed reality allows scientists to interact with overlays seen through the glasses, and these interactions affect the physical lab